The Lorenz reference state of available energy in AN2D and FC2D: physics and algorithms

Notes on the `TraceableEnergetics` code

Contents

1 Available energy and the reference state **2**

2 The rearrangement problem **3**
2.1 Parcels . . . 3
2.2 Slots and costs . . . 4
2.3 Duals and the certificate . . . 5
2.4 When the sort is exact . . . 5

3 Algorithms of the placement **5**
3.1 Sorted rearrangement . . . 6
3.2 Exact rearrangement: the transportation simplex . . . 6
3.3 FC2D: the top pressure . . . 7
3.4 Complexity . . . 7

4 The reference profile **8**

5 Reference heights **8**

6 Available energy **9**

7 Tendency of the reference state **9**

8 Budgets of ACE and APE **10**

9 Diagnostics **11**

10 Verification and cost **11**

11 Decisions **12**

A Not implemented, and future work **13**
A.1 Speedups of the exact rearrangement (deferred) . . . 13
A.2 Other solvers . . . 13
A.3 Other reference states . . . 14
A.4 Resolution and outputs . . . 14

Scope. This note describes how `TraceableEnergetics` computes the Lorenz reference state of a snapshot of AN2D or FC2D, the available energy of each parcel, the tendency of the reference state and the budgets of the available energy (`src/reference_state/`, with the numerical routines in `src/numerics/`). The available energy and its budget are derived in the energy-budget notes (`energy_budget/energy.tex`, "the energy notes"); notation follows

`models/traceable_energetics.tex` ("the main note"). Appendix A lists what was specified or considered but is not implemented.

Notation. ϑ is the conserved variable (`consvar`), q the solute fraction, $h(P,\vartheta,q)$ the specific enthalpy with $\partial h/\partial P = v$ the specific volume, and $\Phi = g(z - z_0)$. The box is $[0, L_x] \times [0, L_z]$ with $M_x \times M_z$ interior cells of size $\Delta x \times \Delta z$; masses are per unit length along y. AN2D has the reference profiles $\rho_R(z)$, $P_R(z)$ with $\partial_z P_R = -g\rho_R$. Reference-state quantities carry a subscript $*$: the reference pressure $P_*(z,t)$, the reference specific volume $v_*(z,t)$ (density $\rho_* = 1/v_*$), the reference height z_* of a parcel, and in AN2D the reduced reference pressure $\pi_* = (P_* - P_R)/\rho_R$.

1 Available energy and the reference state

The static energy of a parcel at height z and pressure P is

$$\Sigma(\vartheta,q,z,P) = \begin{cases} h(P,\vartheta,q) + \Phi(z) & \text{FC2D}, \\ h(P_R(z),\vartheta,q) + \dfrac{P - P_R(z)}{\rho_R(z)} + \Phi(z) & \text{AN2D}, \end{cases} \qquad \Sigma_P = \begin{cases} v & \text{FC2D}, \\ 1/\rho_R & \text{AN2D}. \end{cases} \tag{1}$$

Given a reference pressure $P_*(z,t)$, the available energy (AE) of a parcel is

$$\Pi = \Sigma(\vartheta,q,z,P) - \Sigma\left(\vartheta,q,z_*,P_*(z_*)\right) - \left(P - P_*(z)\right)\Sigma_P(\vartheta,q,z,P) = \Pi_1 + \Pi_2, \tag{2}$$

the sum of the available compressible energy (ACE), the compression from $P_*(z)$ to P at fixed height,

$$\Pi_1 = \Sigma(\vartheta,q,z,P) - \Sigma(\vartheta,q,z,P_*(z)) - \left(P - P_*(z)\right)\Sigma_P(\vartheta,q,z,P) \;\geq 0, \tag{3}$$

zero in AN2D, where Σ is linear in P, and nonnegative in FC2D, where h is concave in P ($\partial_P v < 0$); and the available potential energy (APE), the displacement from z_* to z through the reference state,

$$\Pi_2 = S(z) - S(z_*), \qquad S(z) = \Sigma\left(\vartheta,q,z,P_*(z)\right), \tag{4}$$

with S the static energy of the parcel along the reference profile. The reference profile defines v_* by

$$v_*\,\partial_z P_* = -g \quad \text{(FC2D)}, \qquad \partial_z \pi_* = g\left(\rho_R v_* - 1\right) \quad \text{(AN2D)}. \tag{5}$$

The energy notes require of the reference state:

1. **Level of neutral buoyancy (LNB).** $S'(z_*) = 0$, i.e. $v(P_*(z_*),\vartheta,q) = v_*(z_*)$ (FC2D) or $v(P_R(z_*),\vartheta,q) = v_*(z_*)$ (AN2D), with $N_*^2 = S''(z_*) \geq 0$;
2. **Volume constraint.** The reference state fills the box level by level: for every f,

$$\int f(z_*)\,\Sigma_P(\vartheta,q,z_*,P_*(z_*))\,\mathrm{d}m = \int f(z)\,\mathrm{d}V.$$

These are the stationarity conditions of $\int \Pi\,\mathrm{d}m$ with respect to z_* and P_*.

Potential energies. The potential energy of a state is that of the energetics of the model, the integral of $\rho\left(h - P/\rho + \Phi\right)$, the internal plus gravitational energy (per unit length along y):

$$E_P = \begin{cases} \displaystyle\int \left[\rho\left(h(P,\vartheta,q) + \Phi\right) - P\right]\mathrm{d}A & \text{FC2D}, \\ \displaystyle\int \left[\rho_R\left(h(P_R,\vartheta,q) + \Phi\right) - P_R\right]\mathrm{d}A & \text{AN2D}, \end{cases} \tag{6}$$

where AN2D evaluates the thermodynamics at the reference pressure $P_R(z)$, as its energy does. The **reference energy** E_* is the same integral for the reference state: every fluid element of the actual state, of mass $\mathrm{d}m = \rho\,\mathrm{d}A$ and conserved (ϑ, q), moved adiabatically to its reference height $z_*(\vartheta, q)$, at rest under the reference pressure. Written over the mass, with the reference state filling the box (the volume constraint),

$$E_* = \begin{cases} \displaystyle\int \left[h\left(P_*(z_*), \vartheta, q\right) + \Phi(z_*)\right] \mathrm{d}m - L_x \int_0^{L_z} P_*(z)\,\mathrm{d}z & \text{FC2D}, \\ \displaystyle\int \left[h\left(P_R(z_*), \vartheta, q\right) + \Phi(z_*)\right] \mathrm{d}m - L_x \int_0^{L_z} P_R(z)\,\mathrm{d}z & \text{AN2D}, \end{cases} \tag{7}$$

the integrals in $\mathrm{d}m$ over the fluid (each element at its own z_*), the pressure integrals over the box. In FC2D, P_* is hydrostatic with the density ρ_* of the rearranged fluid, $\partial_z P_* = -g\rho_*$; integrating by parts, $L_x \int_0^{L_z} P_*\,\mathrm{d}z = P_{\mathrm{top}} L_x L_z + \int \Phi(z_*)\,\mathrm{d}m + g z_0 M_{\mathrm{tot}}$, so

$$E_* = \int h\left(P_*(z_*), \vartheta, q\right) \mathrm{d}m - P_{\mathrm{top}} L_x L_z - g z_0 M_{\mathrm{tot}} \qquad \text{(FC2D)}, \tag{8}$$

with $P_{\mathrm{top}} = P_*(L_z)$. In AN2D the mass per unit height of the reference state is that of the model, $L_x \rho_R(z)$, and the pressure term is the same in E_P and E_*.

The **Lorenz reference state** is the rearrangement (the map of the fluid elements to their heights z_*, with the volume constraint, and in FC2D with P_* hydrostatic with it) that minimises E_*. Then $E_P - E_* \geq 0$ (exactly in AN2D, up to an $O(\Delta z^4)$ discretisation difference in FC2D, §2.2) is the potential energy that any adiabatic rearrangement can release, the available potential energy of the whole box in the sense of Lorenz. It is a number for the box, not a sum of parcel energies. When the volume constraint holds,

$$\int \Pi\,\mathrm{d}m = E_P - E_*, \tag{9}$$

since, from (2), the terms of Π at the actual state give E_P plus $\int P\,\mathrm{d}V$ (FC2D; $\int P_R\,\mathrm{d}A$ plus $\int (P - P_R)\,\mathrm{d}A$ in AN2D), those at the reference state E_* plus $L_x \int P_*\,\mathrm{d}z$, and the pressure term $-\int (P - P_*(z))\,\Sigma_P\,\mathrm{d}m = -\int (P - P_*(z))\,\mathrm{d}V$ cancels the rest. For any other profile with the global minimisers z_*, $\int \Pi\,\mathrm{d}m \geq E_P - E_*$ of the Lorenz state (§2.3). The diagnostic `background_potential_energy` is $E_P - \int \Pi\,\mathrm{d}m$, which is E_* when (9) holds. The discrete E_P and E_* are given in §2.2.

The code takes z_* to be the *global* minimiser of S on $[0, L_z]$ (§5): it maximises the parcel's Π, depends on (ϑ, q, t) only, gives $\Pi_2 \geq 0$, and makes the terms in $\partial_t z_*$ drop out of the budgets.

Discretely (§2) the Lorenz reference state is a placement of the parcels into the levels of a column, whose constraints are the discrete volume constraint and whose dual potentials are the discrete P_*.

2 The rearrangement problem

2.1 Parcels

Each interior cell is a parcel $n = 1, \ldots, N$, $N = M_x M_z$, numbered row by row from the bottom. It keeps its mass and conserved properties,

$$m_n = \rho\,\Delta x \Delta z, \qquad \vartheta_n = \frac{(\rho\vartheta)_{ij}}{\rho}, \qquad q_n = \frac{(\rho q)_{ij}}{\rho}, \tag{10}$$

with $\rho = \rho_R(z_j)$ in AN2D and the cell density in FC2D. It also has its own pressure p_n ($P_R(z_n)$ in AN2D; in FC2D the pressure $p(v_n, \vartheta_n, q_n)$ of the equation of state at $v_n = 1/\rho$) and height z_n, the centre of its row.

2.2 Slots and costs

The reference column has $K = M_z$ *slots*, the model rows, numbered from the top, of masses M_k ($\sum_k M_k = M_{\mathrm{tot}}$). A placement $x_{nk} \geq 0$ is the mass of parcel n in slot k; the Lorenz state is the least-cost placement,

$$\min_{x \geq 0} \sum_{n,k} x_{nk}\, c_{nk} \qquad \text{subject to} \qquad \sum_k x_{nk} = m_n \;\; \forall n, \qquad \sum_n x_{nk} = M_k \;\; \forall k, \tag{11}$$

the transportation problem (Hitchcock 1941). It generalises the linear assignment problem of Hieronymus & Nycander (2015) (equal masses, $K = N$) and the minimum-cost flow problem of Su & Ingersoll (2016) (equal masses, $K \ll N$ levels): the parcels have unequal masses here, and may be split between slots. The costs are energies per unit mass measured from the parcel's own state, so that a constant per parcel, which changes neither the placement nor the slot duals, cancels:

- **AN2D.** The slots are the rows at the fixed pressures $P_k = P_R(z_k)$, of masses $M_k = M_x \rho_R(z_k) \Delta x \Delta z$. These are the states anelastic flow can reach ($\nabla \cdot (\rho_R \mathbf{u}) = 0$ preserves $\rho_R \,\mathrm{d}A$). The cost is

  $$c_{nk} = h(P_R(z_k), \vartheta_n, q_n) - h(P_R(z_n), \vartheta_n, q_n) + g\,(z_k - z_n), \tag{12}$$

  zero in the parcel's own row. The discrete energies (6) of the state and of a placement are

  $$E_P = \sum_n \left[m_n \left(h(P_R(z_n), \vartheta_n, q_n) + \Phi(z_n) \right) - P_R(z_n)\, \Delta x \Delta z \right], \tag{13}$$

  $$E_* = \sum_{nk} x_{nk} \left(h(P_R(z_k), \vartheta_n, q_n) + \Phi(z_k) \right) - \sum_k M_x\, P_R(z_k)\, \Delta x \Delta z, \tag{14}$$

  E_P being the integrated potential energy of the energetics. The rows keep their masses, so the pressure terms are equal, and $E_P - E_* = -\sum_{nk} x_{nk} c_{nk}$ exactly. The actual state is the feasible placement "each parcel in its own row", so $E_* \leq E_P$ to round-off for the optimal placement, with equality for a state at rest and stably stratified.

- **FC2D.** The slots are the rows with the masses M_k of the rows of the state, so that a state at rest is its own reference state. The column is hydrostatic in mass, and slot k is at the pressure

  $$P_k = P_{\mathrm{top}} + \frac{g}{L_x} \Big(\sum_{l<k} M_l + \tfrac{1}{2} M_k \Big), \qquad c_{nk} = h(P_k, \vartheta_n, q_n) - h(p_n, \vartheta_n, q_n), \tag{15}$$

  with the top pressure P_{top} set by the condition that the column fills the box (§3.3). Slot k has the volume $V_k = \sum_n x_{nk} v(P_k, \vartheta_n, q_n)$ and the thickness V_k / L_x, stacked from the bottom. Integrating $\rho g z$ by parts in the column, $\sum_k M_k g z_k - \sum_k P_k V_k = -P_{\mathrm{top}} L_x L_z$. The discrete energies (6) of the state and of the column are then

  $$E_P = \sum_n \left[m_n \left(h(p_n, \vartheta_n, q_n) + \Phi(z_n) \right) - p_n\, \Delta x \Delta z \right], \tag{16}$$

  $$\begin{aligned} E_* &= \sum_{nk} x_{nk} \left(h(P_k, \vartheta_n, q_n) + \Phi(z_k) \right) - \sum_k P_k V_k \\ &= \sum_{nk} x_{nk}\, h(P_k, \vartheta_n, q_n) - g z_0 M_{\mathrm{tot}} - P_{\mathrm{top}} L_x L_z, \end{aligned} \tag{17}$$

  the geopotential of the column independent of the placement at fixed P_{top}, and

  $$E_P - E_* = \sum_n \left[m_n g z_n - (p_n - P_{\mathrm{top}}) \Delta x \Delta z \right] - \sum_{nk} x_{nk} c_{nk}. \tag{18}$$

For each placement, P_{top} is the pressure at which its column fills the box, and the Lorenz state is the placement of least E_* (the saddle point of §3.3). For a state at rest, $E_P - E_*$ is the difference between the model's discrete hydrostatic balance and that of the column, $O(\Delta z^4)$ (about $10^{-18} M_{\mathrm{tot}} g L_z$ at $M_z = 8$).

The enthalpy differences are integrals of v by three-point Gauss–Legendre quadrature, $h(P + \Delta P) - h(P) = \int v \,\mathrm{d}P$ (exact for the equation of state of the main note, linear in P). This keeps the relative precision of the float type: the difference of two values of $h \approx 10^6 \,\mathrm{J\,kg^{-1}}$ would carry $2 \times 10^{-10} \,\mathrm{J\,kg^{-1}}$ of round-off in Float64, far above the tolerance of the certificate below.

2.3 Duals and the certificate

The dual of (11) has a potential u_n per parcel and ψ_k per slot. A placement is optimal if and only if there are duals with

$$c_{nk} - u_n - \psi_k \geq 0 \quad \forall n, k, \qquad \text{with equality wherever } x_{nk} > 0 \tag{19}$$

(weak duality and complementary slackness; e.g. Bertsimas & Tsitsiklis 1997, ch. 4). The code checks the first condition for all NK pairs after every placement, the second holding by construction: `certificate_violations` counts the parcels whose least reduced cost $\min_k(c_{nk} - u_n - \psi_k)$ is below $-16(K+1)\,\epsilon\, g L_z$, the round-off of duals that accumulate K cost differences of size up to gL_z (ϵ the machine epsilon of the float type). This pass costs NK evaluations of the cost, threaded and vectorised over the parcels of a row, and is also the pricing of the exact method (§3.2).

The duals are the discrete reference pressure. In AN2D, $S(z_k) = c_{nk} + \pi_*(z_k)$ up to a constant per parcel, so a parcel's optimal slots minimise $c_{nk} - \psi_k$ and $\pi_{*,k} = -\psi_k$ up to a constant; in FC2D the duals give $-\Phi(z_k)$. Dual feasibility with complementary slackness is the discrete LNB condition in global form, and ψ is the multiplier of the slot constraints, the discrete volume constraint. By weak duality, for *any* reference profile, $\int \Pi \,\mathrm{d}m$ with the global minimiser z_* is at least $E_P - E_*$ of the Lorenz state (cf. Tailleux 2013), with equality in AN2D for the optimal duals with z_* restricted to the row centres. The code uses the duals only to certify the placement; the profile comes from the slot volumes (§4).

2.4 When the sort is exact

The equation of state of the main note (Appendix A there) is

$$v = v_0 \left[1 + \alpha_T \left(1 + \gamma(P - P_0)\right)(\vartheta - T_0) - \alpha_q (q - q_0) - \alpha_P (P - P_0) + \tfrac{1}{2}\alpha_{TT}(\vartheta - T_0)^2\right]. \tag{20}$$

If v is additively separable, $v = f(\vartheta, q) + w(P)$, then $h(P, \vartheta, q) = h(P_0, \vartheta, q) + f(\vartheta, q)(P - P_0) + \int_{P_0}^{P} w$, and only $\sum f_n P_k$ depends on the placement. With the parcels ordered by f and the slots by pressure the costs have the Monge property, for which filling the slots in order (the north-west corner rule) is optimal, with unequal masses and split parcels (Hoffman 1963). In FC2D the slot pressures depend on P_{top} but their order does not. So **the sort by specific volume is the exact Lorenz state iff** $\gamma = 0$: compressibility, haline contraction and cabbeling keep v separable, thermobaricity does not. With $\gamma \neq 0$ the order of two parcels can depend on the pressure at which they are compared, and the sort is approximate. In the Boussinesq limit (uniform ρ_R, equal masses) and with $\gamma = 0$ it is the sort of Winters et al. (1995).

3 Algorithms of the placement

The method is a type parameter of the reference state: `SortedRearrangement()` (the default of the constructors) or `ExactRearrangement()` (used by the examples). Both start from the sort, so the two can be compared on the same state.

3.1 Sorted rearrangement

1. The sort key of each parcel is its specific volume at a key pressure, $v(P_{\text{key}}, \vartheta_n, q_n)$: $P_R(L_z/2)$ in AN2D, the mean pressure of the parcels in FC2D. With $\gamma = 0$ every P_{key} gives the same order.

2. The parcels, lightest first, fill the slots from the top, a parcel split across a slot boundary where needed. The entries of the placement form a staircase.

3. FC2D: P_{top} for this placement (§3.3).

4. The duals along the staircase: $\psi_1 = 0$, and across the boundary between slots k and $k+1$ the step of ψ is $c_{a,k+1} - c_{a,k}$ for the parcel a split there; at a degenerate boundary (no split parcel), the middle of the interval allowed by the last parcel of slot k and the first of slot $k+1$, a reproducible choice. Then $u_n = c_{nk} - \psi_k$ on any slot of parcel n.

5. One pass over the pairs: the certificate.

Cost: $O(N \log N)$ for the sort, $O(NK)$ for the certificate.

3.2 Exact rearrangement: the transportation simplex

A transportation simplex (Dantzig 1951), in the network form of Ahuja, Magnanti & Orlin (1993, ch. 11), written in Julia (`src/numerics/transportation.jl`). The network has $N + K$ nodes, the parcels and the slots, with an arc from each parcel to each slot carrying x_{nk}; a basis is a spanning tree of $N + K - 1$ arcs, some of which may carry no mass.

1. **Start.** The sorted placement, completed into a spanning tree: at a degenerate boundary between slots k and $k+1$, the first parcel of slot $k+1$ gets an entry of zero mass in slot k.

2. **Duals.** At the start of each pass, $\psi_1 = 0$ and $u_n + \psi_k = c_{nk}$ on the arcs of the tree, by a breadth-first walk from the first slot, in $O(N + K)$ with the cost of each arc kept.

3. **Pricing.** A pass over all NK pairs (the certificate pass of §2.3) gives each parcel's least reduced cost and its slot. The parcels below $-16(K+1)\epsilon g L_z$ are the candidates, sorted once, most negative first (the lower index first among equals). Each is priced again with the current duals, one cost evaluation, and enters the tree if still negative. Taking the most favourable arc of each node as a candidate is the candidate-queue pricing of large network simplex codes (Bradley, Brown & Graves 1977; see also Grigoriadis 1986). The passes repeat until one finds no candidate: that pass is the certificate.

4. **Pivot.** The entering pair (n, k) closes a cycle in the tree, from n up to the apex (the deepest common ancestor of n and k) and down to k. A mass θ moves round it, $+\theta$ on the arcs oriented as the entering one and $-\theta$ on the others; θ is the least mass of these, so masses stay exactly nonnegative in floating point. The leaving arc is the last arc that blocks θ met going round the cycle from the apex in the direction of the entering arc.

5. **Update of the tree.** Only the subtree that hung from the leaving arc changes: it now hangs from the entering arc, rerooted at the end q of that arc that it holds, and its depths and duals change, in a time of the size of the subtree rather than $O(N + K)$ (as with the threaded indices of Glover, Klingman & Stutz 1974; here with lists of children). With r the reduced cost of the entering pair, the duals of the subtree shift by $u \to u + r$, $\psi \to \psi - r$ if q is the parcel, and $u \to u - r$, $\psi \to \psi + r$ if q is the slot. Step 2 recomputes the duals from the costs at each pass, so the certificate carries the round-off of the costs only.

The tree is rooted at the first slot and kept *strongly feasible* (Cunningham 1976): every arc of zero mass goes from a parcel up to its parent slot, which the completed staircase satisfies and the leaving rule preserves. A pivot that moves no mass, frequent from the degenerate staircase, then still makes progress, and the method cannot cycle. The optimum is global: (11) is a linear programme, and the certificate proves it. The pivots give the decrease of the cost, $\sum \theta\,|c_{nk} - u_n - \psi_k|$, by which the sort underestimates $E_P - E_*$ (`sorted_energy_excess`).

The number of passes grows with how far the sort is from the optimum: one where it is optimal, a few where it fails for a few nearly tied parcels, about K where every parcel is misplaced, as each pass moves a misplaced parcel about one slot nearer its level. The passes are capped at $10K$ (`passes_per_slot`), which only guards against round-off; a placement cut short fails its certificate.

3.3 FC2D: the top pressure

For a placement x, let $G(x, P_{\mathrm{top}}) = \sum x_{nk} c_{nk}(P_{\mathrm{top}}) - P_{\mathrm{top}} L_x L_z$, the energy of the column up to terms independent of both. Then $\partial G/\partial P_{\mathrm{top}} = \sum x_{nk} v(P_k, \vartheta_n, q_n) - L_x L_z$ is the volume excess and $\partial^2 G/\partial P_{\mathrm{top}}^2 = \sum x_{nk} \partial_P v < 0$: G is concave in P_{top}, with its maximum where the column fills the box. The reference state is the saddle point

$$\min_x \max_{P_{\mathrm{top}}} G(x, P_{\mathrm{top}}) = \max_{P_{\mathrm{top}}} \min_x G(x, P_{\mathrm{top}}) \tag{21}$$

(G is linear in x over a compact convex set and concave in P_{top}; Sion's minimax theorem), the discrete counterpart of the statement of the energy notes that $\int \Pi \,\mathrm{d}m$ is a minimum with respect to P_*, and of the free-surface optimal transport of Egan et al. (2025).

- P_{top} **at fixed placement** (`top_pressure!`): Newton's method on the volume residual $R(P_{\mathrm{top}}) = \sum x_{nk} v(P_k) - L_x L_z$, with $R' = \sum x_{nk} \partial_P v(P_k)$, from the mean over the top row of $p_n - g\Delta z/(2v_n)$. It stops when $|R| \le 16\epsilon L_x L_z$ or $|R|$ no longer decreases.
- **Sorted method:** the placement does not depend on P_{top} when $\gamma = 0$, so P_{top} is computed once.
- **Exact method:** from the sorted placement, P_{top} at fixed placement and the simplex at fixed P_{top} (from the previous tree) alternate until a simplex moves no mass, at most 20 times (`max_top_iterations`); the last placement is then only priced, and its certificate fails. At the end both conditions hold: the placement is optimal at P_{top}, and P_{top} fills the box with it, which is the saddle point.

3.4 Complexity

Of the problem. (11) is a transportation problem, an uncapacitated minimum-cost flow problem on $N+K$ nodes and NK arcs, so it is solvable in strongly polynomial time, in a number of operations bounded by a polynomial in N and K alone (Tardos 1985). Orlin's (1993) algorithm solves uncapacitated minimum-cost flow on n nodes and m arcs in $O(n \log n\,(m + n \log n))$, here $O(N^2 K \log N)$; Kleinschmidt & Schannath (1995) give a strongly polynomial algorithm for the transportation problem itself. With few demand points, $K \ll N$, Tokuyama & Nakano (1995) solve it in expected $O(KN)$, which is optimal, when $N = \Omega(K^{4+\varepsilon})$ and the ratio of the largest to the smallest parcel mass is polynomial in N; our grids ($N = 2048$, $K = 32$) are far from that regime. Without thermobaricity the costs are Monge (§2.4) and the problem is a sort, $O(N \log N)$. Equal masses with $K = N$ is linear assignment, $O(N^3)$.

Of the algorithm used. The primal network simplex with a strongly feasible tree terminates (Cunningham 1976), but no polynomial bound on the number of its pivots is known for pricing rules such as ours, the most negative reduced cost of each parcel; with the closely related rule of Dantzig, the most negative reduced cost overall, network simplex methods take exponentially many pivots on some networks (Zadeh 1973). Polynomial pivot rules exist (Orlin 1997). The observed cost is (passes) $\times NK$ cost evaluations, plus small per-pivot terms (§3.2): one pass where the sort is optimal, a few where it fails for a few nearly tied parcels, and at most about K in the cases tested (1.0 to $1.3K$ where every parcel is misplaced, as each pass moves a misplaced parcel about one slot nearer its level), i.e. $O(NK^2)$ evaluations. This is an observation on our states, not a theorem. The certificate proves each result optimal however it was found, so a solver with a guaranteed bound could replace or back the simplex without changing anything else (Appendix A).

4 The reference profile

The profile is built from the slot-mean specific volume and compressibility of the parcels of each slot,

$$v_{*,k} = \frac{1}{M_k} \sum_n x_{nk}\, v(P_k, \vartheta_n, q_n), \qquad b_k = \frac{1}{M_k} \sum_n x_{nk}\, \partial_P v(P_k, \vartheta_n, q_n), \tag{22}$$

which change continuously with the state, unlike the duals (§11). The pressure is interpolated, and v_* derived from the interpolant, so that Π, v_*, N_*^2 and the LNB residual come from one profile and z_* is an exact minimiser of the S in Π.

- **AN2D.** P_R is the C^1 cubic Hermite interpolant through the cell centres with the slopes $-g\rho_R$. π_* is a cubic Hermite interpolant with nodes at the row centres and both walls. At the row centres its slopes are $g(\rho_R v_{*,k} - 1)$ and its values follow from $\partial_z \pi_* = -v_* \partial_z P_R - g$ with v_* linear in P_R between the rows (the trapezoidal rule in the pressure),

$$\pi_{*,c} - \pi_{*,c-1} = -\tfrac{1}{2}(v_{*,c-1} + v_{*,c})\,(P_R(z_c) - P_R(z_{c-1})) - g(z_c - z_{c-1}), \tag{23}$$

  exact when v is linear in P, so that the static energy of a neutral fluid is flat at the nodes. Between the end row e and the wall, v_* follows the compression of the parcels of the end slot, $v_* = v_{*,e} + b_e(P_R - P_R(z_e))$, which gives the wall nodes $\pi_* = \pi_{*,e} - v_{*,e}\Delta P - \frac{1}{2} b_e \Delta P^2 - g(z - z_e)$, $\Delta P = P_R(z) - P_R(z_e)$. (A constant v_* there lowered the static energy of these parcels towards the walls by $\frac{1}{2}|b_e|(g\rho_R \Delta z/2)^2$ and made a wall the reference height of every parcel of a neutral fluid.) π_* is defined up to a constant, which changes neither z_* nor Π; its mean over the mass of the rows is zero. On the model rows, $P_* = P_R + \rho_R \pi_*$ and $v_* = (\partial_z \pi_*/g + 1)/\rho_R$.

- **FC2D.** The slots are stacked from the bottom with the thicknesses V_k/L_x; edge e has the pressure P_{top} plus g/L_x times the mass above it. P_* is the monotone C^1 cubic (PCHIP: Fritsch–Carlson slopes with Brodlie's weights, three-point end slopes) through the edge pressures, with the secants $-g/v_{*,k}$, so that the slot density M_k/V_k is exactly the mean of $-\partial_z P_*/g$ across the slot. On the model rows, $v_* = -g/\partial_z P_*$. The column fills the box to the round-off of P_{top} (`volume_residual`).

5 Reference heights

The reference height of a parcel is the global minimiser of its static energy along the profile,

$$z_* = \arg\min_{0 \le z \le L_z} S(z), \qquad r(z) = -S'(z) = \begin{cases} g\rho_R\,(v(P_R(z), \vartheta, q) - v_*(z)) & \text{AN2D}, \\ g\,(\rho_* v(P_*(z), \vartheta, q) - 1) & \text{FC2D}, \end{cases} \tag{24}$$

with r the LNB residual, $-(v\,\partial_z P_R + g + \partial_z \pi_*)$ in AN2D and $-(v\,\partial_z P_* + g)$ in FC2D before (5) is used ($r > 0$: the parcel is lighter than the reference state at z), and $N_*^2 = S'' = \partial_P v\,(\partial_z P)^2 + v\,\partial_z^2 P + \partial_z^2 \pi_*$ (no π_* in FC2D), which depends on the parcel through (ϑ, q) (Tailleux 2013, Appendix A). S is measured from the parcel's own pressure and height, with the enthalpy differences by quadrature. For each parcel, by the model's manager:

1. **Coarse search**, vectorised over the parcels of a row: S at the levels of the profile, both walls and the row centres in AN2D, both walls and the interior slot edges in FC2D, and the lowest (the lowest level among equal values).

2. **Refinement** next to that level: if r changes sign towards a neighbouring level, its root, found by the bracketing method of Alefeld, Potra & Shi (1995) to within $\sqrt{\epsilon}\,L_z$ (which leaves S at round-off), is kept if S is lower there. A wall is the minimum where r has the outward sign.

3. **The parcel's own height** is the reference height if S is lower still there (in FC2D it is not a level, and in a well-mixed fluid, where S is flat to round-off, it can be the lowest point found).

A local minimum next to another level is not refined; it can be lower than the one found only if the two differ by less than the change of S between two levels. Multiple minima arise only through thermobaricity. The global minimiser (rather than the nearest root of Tailleux 2013, §3.1) makes z_* independent of the parcel's height, so Π does not jump when a parcel crosses a local maximum of S; $S(z_*)$ is continuous in (ϑ, q, t) even where z_* jumps between two minima. The number of parcels at a wall is a diagnostic (`wall_minima`).

For a parcel in a slot that holds a mixture, z_* can differ from its placement by up to a slot thickness, as v_* is a slot average: z_* satisfies the LNB condition, which the local budgets need, and the placement the volume constraint.

6 Available energy

On the cells: $\Pi_2 = S(z_n) - S(z_*) \geq 0$, with both values measured from the parcel's own state as in the search, so that $\Pi_2 \geq 0$ holds exactly; $\Pi_1 = 0$ in AN2D; in FC2D

$$\Pi_1 = h(p) - h(P_*(z)) - (p - P_*(z))\,v(p) = -\int_{P_*(z)}^{p} (s - P_*(z))\,\partial_P v(s, \vartheta, q)\,\mathrm{d}s \;\geq 0, \qquad (25)$$

by quadrature, with $P_*(z)$ at the centre of the parcel's row. Their integrals are $\sum_n m_n \Pi_n$; the background potential energy is $E_P - \sum_n m_n \Pi_n$, with E_P the integrated potential energy of the energetics.

7 Tendency of the reference state

The budgets need $\partial_t P_*$ on the model rows and at each z_*. The Lorenz state depends only on the distribution of mass over (ϑ, q), which advection does not change, so P_* changes only through the diabatic rates (Winters et al. 1995; Winters & D'Asaro 1996)

$$\dot\vartheta = \frac{\sigma_\vartheta - \nabla\cdot\mathbf{J}_\vartheta}{\rho}, \qquad \dot q = -\frac{\nabla\cdot\mathbf{J}_q}{\rho}, \qquad \dot v = \partial_\vartheta v\,\dot\vartheta + \partial_q v\,\dot q \;\; \text{(at fixed pressure)}, \qquad (26)$$

which the model provides ($\rho = \rho_R$ in AN2D). The tendency is computed from a single snapshot (`reference_tendency!`).

Derivation. Follow a parcel at its reference level and differentiate the LNB condition. In FC2D, in the mass coordinate m (the mass above a level), $P_*(m,t) = P_{\mathrm{top}}(t) + gm/L_x$, and with $N_*^2 = g\rho_*^2(v_*\partial_z v_* + gv_P)$,

$$\frac{N_*^2}{g\rho_*^2 L_x}\,\frac{Dm_*}{Dt} = \partial_t v_*|_m - \dot{v} - v_P\,\partial_t P_{\mathrm{top}}. \tag{27}$$

The mass above a level stays the same, so the mass-weighted mean of Dm_*/Dt over the parcels at a level vanishes:

$$\partial_t v_*|_m = \frac{\langle(\dot{v} + v_P\,\partial_t P_{\mathrm{top}})/N_*^2\rangle}{\langle 1/N_*^2\rangle}, \qquad \int_0^{M_{\mathrm{tot}}} \partial_t v_*|_m \,\mathrm{d}m = 0, \tag{28}$$

the second because the box stays full, which fixes $\partial_t P_{\mathrm{top}}$. In AN2D the levels are fixed and there is no lid: $\partial_t v_* = \langle \dot{v}/N_*^2\rangle/\langle 1/N_*^2\rangle$, $\partial_z(\partial_t \pi_*) = g\rho_R \partial_t v_*$, $\partial_t P_* = \rho_R \partial_t \pi_*$. The weights $1/N_*^2$ differ between the parcels of a level only through v_P (thermobaricity): a weakly stratified parcel moves further for a given change of volume and displaces more of its neighbours. The plain mean, which a derivative at fixed placement gives, is exact only for a separable v.

Discrete form. The parcels at a level are those of a slot, with the masses of the placement (entries of zero mass skipped). For parcel n in slot k, with $\dot{v}$ and v_P at (P_k, ϑ_n, q_n),

$$N_{nk}^2 = g\rho_k\left(\partial_z v_*(z_k) + g\rho_k v_{P,nk}\right), \qquad w_{nk} = \frac{x_{nk}}{N_{nk}^2}, \qquad \langle a\rangle_k = \frac{\sum_n w_{nk} a_{nk}}{\sum_n w_{nk}}, \tag{29}$$

with $\rho_k = \rho_R(z_k)$ (AN2D) or $1/v_{*,k}$ (FC2D) and $\partial_z v_*$ from the profile. A slot with $N_{nk}^2 \le 0$ for one of its parcels takes the plain weights $w_{nk} = x_{nk}$ and is counted (`plain_weight_slots`).

- **AN2D:** $\partial_t v_{*,k} = \langle \dot{v}\rangle_k$, and $\partial_t \pi_*$ by the rule of π_*: slopes $g\rho_R \partial_t v_{*,k}$, values by the trapezoidal rule in P_R, zero mean over the mass of the rows. Between the end rows and the walls only $\partial_t v_{*,e}$ is kept: $\partial_t b_e$ needs the mixed derivatives $\partial^2 v/\partial P\partial\vartheta$, $\partial^2 v/\partial P\partial q$, which the equation of state does not provide. The neglected term of $\partial_t \pi_*$, $-\frac{1}{2}\partial_t b_e \Delta P^2$, is at most 2.6×10^{-3} of $\max|\partial_t \pi_*|$ for $\gamma = 3\times10^{-4}\,\mathrm{Pa}^{-1}$ at $M_z = 16$ (`tools/reference_tendency_study.jl`).
- **FC2D:**

$$\partial_t P_{\mathrm{top}} = -\frac{\sum_k M_k \langle \dot{v}\rangle_k}{\sum_k M_k \langle v_P\rangle_k}, \qquad \partial_t v_{*,k} = \langle \dot{v}\rangle_k + \langle v_P\rangle_k\,\partial_t P_{\mathrm{top}}, \tag{30}$$

  at fixed slot masses. The slot edges move at $\partial_t z_e = L_x^{-1}\sum_{k\ \text{below}\ e} M_k \partial_t v_{*,k}$, zero at both walls, and their pressures at $\partial_t P_{\mathrm{top}}$. $\partial_t P_*$ at fixed height is the derivative of the monotone cubic P_* with respect to these motions of its nodes, by dual numbers (value $+\,\varepsilon\times$ tendency for the heights, pressures and secants of the edges, through the same slopes and cubic as P_*); at an edge it is $\partial_t P_{\mathrm{top}} + g\rho_* \partial_t z_e$.

$\int R_\Pi \,\mathrm{d}V$ (the exchange of the terms in $\partial_t P_*$ with the rest) vanishes to the resolution of the slots, as the reference heights of the parcels of a mixed slot differ from their placement by up to a slot thickness.

8 Budgets of ACE and APE

The budgets of $\rho\Pi_1$ and $\rho\Pi_2$ of the energy notes are diagnosed as the budget of ρP of the energetics is, from the fluxes of the tendencies and the operators of the model (`src/reference_state/budget.jl`). The derivatives Σ_ϑ, Σ_q of the static energy are evaluated at three states of a parcel: the state itself (z,p); the intermediate state $(z, P_*(z))$, marked with a tilde ($\tilde{\Sigma}_\vartheta$); and the reference state $(z_*, P_*(z_*))$, marked with a star (Σ_ϑ^*). With $\Delta_1\Sigma_\vartheta = \Sigma_\vartheta - \tilde{\Sigma}_\vartheta$

(zero in AN2D), $\Delta_2\Sigma_\vartheta = \tilde{\Sigma}_\vartheta - \Sigma^*_\vartheta$ (likewise for q, by quadrature of $\partial_\vartheta v$, $\partial_q v$ in the pressure), at a fixed reference state

$$\partial_t(\rho\Pi_i) = B_i\,\partial_t\rho + \Delta_i\Sigma_\vartheta\,\partial_t(\rho\vartheta) + \Delta_i\Sigma_q\,\partial_t(\rho q), \tag{31}$$

with

$$B_1 = \Pi_1 + (p - P_*)\Sigma_P - \vartheta\Delta_1\Sigma_\vartheta - q\Delta_1\Sigma_q, \qquad B_2 = \Pi_2 - \vartheta\Delta_2\Sigma_\vartheta - q\Delta_2\Sigma_q. \tag{32}$$

In Bernoulli form, with the reversible flux $\mathbf{J}^{\mathrm{rev}} = \rho\mathbf{u}\,\Im(B) + \rho\vartheta\mathbf{u}\,\Im(\Delta\Sigma_\vartheta) + \rho q\mathbf{u}\,\Im(\Delta\Sigma_q)$, the irreversible flux $\mathbf{J}^{\mathrm{irr}} = \Im(\Delta\Sigma_\vartheta)\mathbf{J}_\vartheta + \Im(\Delta\Sigma_q)\mathbf{J}_q$, the conversion $C_B = -\rho\mathbf{u}\cdot\nabla B - \rho\vartheta\mathbf{u}\cdot\nabla\Delta\Sigma_\vartheta - \rho q\mathbf{u}\cdot\nabla\Delta\Sigma_q$ to kinetic energy, and the heatings $Q_\vartheta = -\mathbf{J}_\vartheta\cdot\nabla\Sigma_\vartheta$, $Q_q = -\mathbf{J}_q\cdot\nabla\Sigma_q$, $Q = \Sigma_\vartheta\sigma_\vartheta$ at each state, the product rule of the operators, $\delta(\Im(a)b) = a\,\delta b + \Im(b\,\delta a)$, closes at each cell to round-off (in AN2D up to the divergence of $\rho_R\mathbf{u}$)

$$\partial_t(\rho\Pi_1) = -\nabla\cdot(\mathbf{J}^{\mathrm{rev}}_{\mathrm{ACE}} + \mathbf{J}^{\mathrm{irr}}_{\mathrm{ACE}}) - C_{\mathrm{ACE}} - (Q_\vartheta - \tilde{Q}_\vartheta) - (Q_q - \tilde{Q}_q) + Q - \tilde{Q} + R_{\Pi_1}, \tag{33}$$

$$\partial_t(\rho\Pi_2) = -\nabla\cdot(\mathbf{J}^{\mathrm{rev}}_{\mathrm{APE}} + \mathbf{J}^{\mathrm{irr}}_{\mathrm{APE}}) - C_{\mathrm{APE}} - (\tilde{Q}_\vartheta - Q^*_\vartheta) - (\tilde{Q}_q - Q^*_q) + \tilde{Q} - Q^* + R_{\Pi_2}, \tag{34}$$

with the terms in $\partial_t P_*$ of the energy notes, $R_{\Pi_1} = \rho\,(\Sigma_P(z,p) - \Sigma_P(z,P_*(z)))\,\partial_t P_*(z)$ (zero in AN2D) and $R_{\Pi_2} = \rho\,\Sigma_P(z,P_*(z))\,\partial_t P_*(z) - \rho\,\Sigma_P(z_*,P_*(z_*))\,\partial_t P_*(z_*)$. The residuals of both budgets are diagnostics. So are the continuous forms of the energy notes, $\mathbf{J}^{\mathrm{rev}}_{\mathrm{ACE}} = \rho\Pi_1\mathbf{u} + \mathbf{J}^*_C$, $C_{\mathrm{ACE}} = C^*_{b,1} + C^*_D - \nabla\cdot\mathbf{J}^*_C$, $C_{\mathrm{APE}} = C^*_{b,2}$, with $\mathbf{J}^*_C = (p - P_*)\mathbf{u}$, the buoyancy conversions $C^*_{b,1} = \rho g\rho_*(v - \tilde{v})w$, $C^*_{b,2} = \rho g\rho_*(\tilde{v} - v_*)w$ ($\tilde{v} = v(P_*(z))$; in AN2D $C^*_{b,1} = 0$, $C^*_{b,2} = \rho_R g\rho_R(v(P_R) - v_*)w$) and the pressure dilatation $C^*_D = (p - P_*)\nabla\cdot\mathbf{u}$ (zero in AN2D), with the residuals of these manipulations, which converge with the grid. The conversion C from P to K of the energetics is $C_{\mathrm{ACE}} + C_{\mathrm{APE}} + C_{\mathrm{bg}}$, with the background conversion C_{bg} zero in the continuous equations.

9 Diagnostics

The reference state is computed once per diagnostics session (`diagnostics(model; reference_state)`), and its diagnostics are entries of the book:

- **On the cells:** z_*; ACE, APE, AE; the potentials at the intermediate and reference states and their differences; B_1, B_2; the fluxes, their divergences, conversions, heatings and buoyancy terms of the budgets; the diabatic rates; $\partial_t P_*(z_*)$; R_{Π_1}, R_{Π_2}; $\partial_t(\rho\Pi_1)$, $\partial_t(\rho\Pi_2)$ and the residuals.
- **Profiles on the z axis:** P_*, v_*, $\partial_t P_*$.
- **Numbers:** $\int\rho\Pi_1$, $\int\rho\Pi_2$, $\int\rho\Pi$; the background potential energy; the certificate violations; the pivots, pricing passes and `sorted_energy_excess`; the parcels at a wall; the slots with plain weights; in FC2D P_{top}, its tendency, the volume residual and the iterations of the top pressure.

10 Verification and cost

The tests (`test/reference_state.jl`, `transportation.jl`, `exact_rearrangement.jl`, `reference_tendency.jl`, `available_energy_budget.jl`) check, on small grids and for both models:

- **Exact cases:** a state at rest is its own reference state (each parcel in its own row, $E_* = E_P$, Π at round-off); a fully mixed fluid has $E_P - E_*$ and Π at round-off (AN2D, uniform ρ_R) or $O(\Delta z^4)$; the analytic $E_P - E_*$ of a rearranged column with a separable equation of state; brute force over all assignments of 8 parcels, separable and thermobaric.

- **The simplex:** on random cost matrices with unequal masses, feasibility and optimality checked independently of the solver; the linear assignment case against brute force; degenerate problems, including one on which a simplex without a strongly feasible tree can cycle; with $\gamma = 0$ the exact method moves no mass; with strong thermobaricity it passes the certificate the sort fails and releases more energy.

- **Properties:** mass conservation of every parcel and slot; the column fills the box (FC2D); invariance under a periodic shift and under the origin of the geopotential; weak duality; $r(z_*) = 0$ and $N_*^2 \geq 0$ at interior minima, $\Pi_1, \Pi_2 \geq 0$; the same results on serial and threaded managers; no state-sized allocations.

- **Tendency:** with $\gamma = 0$ it matches centred differences of the reference state under $\pm\varepsilon$ times the diabatic rates; the tendencies of $\rho\Pi_1$, $\rho\Pi_2$ with the R terms match differences with the reference state recomputed; without diffusion it vanishes; $\sum_k M_k \partial_t v_{*,k} = 0$ (FC2D); $|\int R_\Pi| / \int |R_\Pi|$ converges at first order (2×10^{-2} to 2×10^{-3} from $M_z = 8$ to 64).

- **Weights under thermobaricity:** two density-compensated families mixed in every row, one heated: the N_*^{-2}-weighted mean is 26%, 8.5% and 5.7% from centred differences of the exact rearrangement at $K = 20$, 40, 80, the plain mean 35% (`tools/exact_rearrangement_study.jl` at full size).

- **Budgets:** the budgets of ACE and APE close at each cell to round-off, with and without the R terms.

Measured (`tools/exact_rearrangement_study.jl`, 2026-10-27 and 2026-10-31).

- Thermohaline lock exchange ($\gamma = 10^{-7}\,\mathrm{Pa}^{-1}$, 64×32): the sort fails the certificate for at most 3 nearly tied parcels, fixed by 2 or 3 pivots; the exact method changes $E_P - E_*$ by at most 9×10^{-10} of it and P_* by at most $2 \times 10^{-8}\,\mathrm{Pa}$, with the same roughness in time. There the sort is the Lorenz state to round-off. `reference_state!` takes 0.4–0.6 ms.

- Every parcel misplaced by the sort (two compensated families, $\gamma = 1.3 \times 10^{-4}\,\mathrm{Pa}^{-1}$): 315, 1114 and 7405 pivots in 21, 38 and 104 passes at $K = 20$, 40, 80; 50 ms at $K = 80$, $N = 1600$, of which pricing is two thirds.

- Float32 sees no thermobaric reordering for γ up to $10^{-3}\,\mathrm{Pa}^{-1}$, where Float64 makes 95 pivots that move mass: the reduced costs are below its tolerance. The exact method needs Float64.

11 Decisions

Taken with the user, and governing the implementation:

- **Exact solver:** a transportation simplex written in Julia, generic in the float type, started from the sorted placement at each snapshot (2026-10-27). The sort and the exact method stay separate methods, to be compared; the sort is the default, the examples use the exact one. The tree is updated in place after each pivot (2026-10-31).

- **Lorenz profile** from the slot-mean specific volumes, in both models; the duals certify the placement only (2026-10-27). The dual profile changes abruptly when the parcel split between two slots changes, and a split parcel has the same static energy at both slots along it, so its z_* jumps between them; both made the budgets rough in time and space (the generation by two orders of magnitude in the lock exchange).

- **Reference height:** the global minimiser of S, not the nearest root of Tailleux (2013) nor the placement.
- **AN2D beyond the end rows:** v_* follows the compression of the parcels of the end slot (2026-10-26); $\partial_t b_e$ is neglected, with its size measured (2026-10-27).
- **Tendency of the reference state** from the diabatic tendencies with N_*^{-2} weights, valid for any equation of state.
- **FC2D $\partial_t P_*$ at fixed height** as the derivative of the monotone cubic with respect to its nodes, by dual numbers (2026-10-27). A hand-written tangent agreed to round-off at the same cost; applying $\partial_t P_* = \partial_t P_{\mathrm{top}} + g\rho_* \partial_t z|_m$ between the edges with $\partial_t z|_m$ linear is not the derivative of the P_* of Π (off by 0.4 to 0.7%).
- **FC2D reference heights in an exactly neutral layer** (2026-10-26, no change): identical parcels with $N_*^2 = 0$ have an S flat up to the wiggles of the monotone cubic, about $10^{-10}\,\mathrm{J\,kg^{-1}}$ at $M_z = 32$; the coarse search then takes the lowest tied level, the bottom wall. Their Π_2 is about $6 \times 10^{-10}\,\mathrm{J\,kg^{-1}}$, and the ties vanish after the first time step.

A Not implemented, and future work

A.1 Speedups of the exact rearrangement (deferred)

- **Banded pricing:** price the pairs within $|k - k_n| \le B$ slots of the sorted slot before the full pass. Pricing is two thirds of the time where every parcel is misplaced, so this is the next lever there.
- **Warm start from the previous snapshot.** The parcels are the cells, so after advection the previous placement has no direct relation to the new state; in AN2D it stays feasible (fixed masses), in FC2D the flows would need recomputing.
- **Warm start of the sort** from the previous order (insertion sort), with a fallback to the full sort when the flow overturns. At most about 1 ms of 16 ms per call at 256×128.
- **A starting interval for the Hermite search** in the refinement of z_*, which evaluates the profile in one bracket. Estimated 20–40% of an evaluation; not measured.

A.2 Other solvers

- **Greedy filling with repair.** Fill the slots from the top with the parcels of largest $v(P_k)$ at each slot pressure (Huang 2005; Wong et al. 2016; the discrete counterpart of Saenz et al. 2015), or bottom-up, or by divide and conquer at the pressure of the boundary that halves the mass (Stansifer et al. 2017), then repair by pairwise exchanges within a band of slots. No optimality guarantee: unrepaired heuristics can give $E_P - E_* < 0$ (Harris & Tailleux 2018), and 2-opt can stall (Hieronymus & Nycander 2015). Not needed while the exact method is fast enough.
- **Histogram reduction** (Tseng & Ferziger 2001; Saenz et al. 2015): bin the parcels in (ϑ, q) and solve for the bins. Averaging (ϑ, q) in a bin biases the energy under a nonlinear equation of state, and a resting stratified state no longer has zero available energy exactly; worthwhile only for much larger grids or long series.
- **Dual ascent:** maximise the concave piecewise-linear dual $D(\psi) = \sum_k M_k \psi_k + \sum_n m_n \min_k(c_{nk} - \psi_k)$ in K variables, whose supergradient is the volume residual of each slot, by bundle or subgradient methods or entropic smoothing; the discrete counterpart of

the semi-discrete optimal transport of Egan et al. (2025). Ties need a small flow among the tied slots.

- **Linear assignment** (Hungarian, Jonker–Volgenant) for equal masses with $K = N$: $O(N^3)$, and 10^3–10^5 times slower than minimum-cost flow on the problem of Su & Ingersoll (2016).

- **A guaranteed bound** (§3.4): a polynomial pivot rule for the network simplex (Orlin 1997), or a strongly polynomial minimum-cost flow algorithm (Orlin 1993), with the same certificate. Likely slower on our states, where the sort is nearly optimal and the simplex needs 1 to 4 passes; worth it only if states appear on which the number of passes grows beyond about K.

A.3 Other reference states

- **Horizontal-mean reference state:** P_* hydrostatic with the row mean of the density (FC2D, by the model's discrete hydrostatic rule, anchored at one level) or of v at P_R (AN2D), with z_* by the same LNB search. Not a rearrangement: $\int R_\Pi \,\mathrm{d}V \neq 0$ and $\int \Pi \,\mathrm{d}m$ exceeds $E_P - E_*$ of the Lorenz state. Cheap, and local. Its tendency would follow from the model's full tendencies, as it changes under advection too. It was implemented with the first version of the Lorenz state and removed with the user (2026-10-26), as it is not a goal of the project.

- **Eddy-size constrained reference state** (Su & Ingersoll 2016): forbid moves beyond a horizontal distance, which turns the problem back into a sparse assignment problem.

A.4 Resolution and outputs

- **Slot resolution other than the rows** ($K \neq M_z$; s slots per row in AN2D, equal-mass slots in FC2D), which the first design made a required input. With $K \neq M_z$ the actual state is no longer a placement with exactly the model's energy, and a state at rest gets $E_P - E_* = O(\Delta z^2)$.

- **Refinement of every near-lowest local minimum** of S, with a count of the parcels with two minima within a tolerance, where z_* can jump.

- **Placement height** $\hat{z}_n = \sum_k x_{nk} z_k / m_n$ (or the sub-row height in sort order, the analogue of the Winters et al. height), as a check of z_*, and the reference static energy and N_*^2 per parcel as outputs.

- $\partial_t P_*$ **from differences of successive snapshots,** as a diagnostic. The tendency from the diabatic rates excludes the changes of the reference state caused by the discretisation of advection, and the differences include them, so their difference measures spurious mixing locally, the counterpart of the background potential energy of Ilıcak et al. (2012). Snapshot intervals must be long compared with $\Delta x/|\mathbf{u}|$, short compared with the diabatic evolution of P_*.

- **FC2D slot masses under advection:** the tendency holds the slot masses (the row masses of the state) fixed, although advection changes them.

- $\partial_t b_e$ **beyond the AN2D end rows,** once the equation of state provides the mixed second derivatives.

References

- Ahuja, Magnanti & Orlin (1993), *Network Flows: Theory, Algorithms, and Applications*, Prentice Hall, ISBN 978-0-13-617549-0 (no DOI).
- Alefeld, Potra & Shi (1995), Algorithm 748: enclosing zeros of continuous functions, ACM Trans. Math. Softw. 21, 327–344, doi:10.1145/210089.210111.
- Bertsimas & Tsitsiklis (1997), *Introduction to Linear Optimization*, Athena Scientific, ISBN 978-1-886529-19-9 (no DOI).
- Bradley, Brown & Graves (1977), Design and implementation of large scale primal transshipment algorithms, Management Sci. 24, 1–34, doi:10.1287/mnsc.24.1.1.
- Cunningham (1976), A network simplex method, Math. Programming 11, 105–116, doi:10.1007/BF01580379.
- Dantzig (1951), Application of the simplex method to a transportation problem, in T. C. Koopmans (ed.), *Activity Analysis of Production and Allocation*, Cowles Commission Monograph 13, Wiley, 359–373 (no DOI; Cowles Foundation).
- Egan, Methven, Bourne & Cullen (2025), A definition of the background state of the atmosphere using optimal transport, arXiv:2504.11141.
- Glover, Klingman & Stutz (1974), Augmented threaded index method for network optimization, INFOR 12, 293–298, doi:10.1080/03155986.1974.11731583.
- Grigoriadis (1986), An efficient implementation of the network simplex method, Math. Programming Study 26, 83–111, doi:10.1007/BFb0121089.
- Harris & Tailleux (2018), Assessment of algorithms for computing moist available potential energy, QJRMS 144, 1501–1510.
- Hieronymus & Nycander (2015), Finding the minimum potential energy state by adiabatic parcel rearrangements with a nonlinear equation of state: an exact solution in polynomial time, JPO 45, 1843–1857, doi:10.1175/JPO-D-14-0174.1.
- Hitchcock (1941), The distribution of a product from several sources to numerous localities, J. Math. Phys. 20, 224–230, doi:10.1002/sapm1941201224.
- Hoffman (1963), On simple linear programming problems, in *Convexity*, Proc. Symp. Pure Math. 7, AMS, 317–327, doi:10.1090/pspum/007/0157778.
- Huang (2005), Available potential energy in the world's oceans, J. Mar. Res. 63, 141–158.
- Ilıcak, Adcroft, Griffies & Hallberg (2012), Spurious dianeutral mixing and the role of momentum closure, Ocean Modelling 45–46, 37–58.
- Kleinschmidt & Schannath (1995), A strongly polynomial algorithm for the transportation problem, Math. Programming 68, 1–13, doi:10.1007/BF01585755.
- Orlin (1993), A faster strongly polynomial minimum cost flow algorithm, Oper. Res. 41, 338–350, doi:10.1287/opre.41.2.338.
- Orlin (1997), A polynomial time primal network simplex algorithm for minimum cost flows, Math. Programming 78, 109–129, doi:10.1007/BF02614365.
- Saenz, Tailleux, Butler, Hughes & Oliver (2015), Estimating Lorenz's reference state in an ocean with a nonlinear equation of state for seawater, JPO 45, 1242–1257.
- Stansifer, O'Gorman & Holt (2017), Accurate computation of moist available potential energy with the Munkres algorithm, QJRMS 143, 288–292.

- Su & Ingersoll (2016), On the minimum potential energy state and the eddy size-constrained APE density, JPO 46, 2663–2674, doi:10.1175/JPO-D-16-0074.1.
- Tardos (1985), A strongly polynomial minimum cost circulation algorithm, Combinatorica 5, 247–255, doi:10.1007/BF02579369.
- Tailleux (2013), Available potential energy density for a multicomponent Boussinesq fluid with arbitrary nonlinear equation of state, JFM 735, 499–518.
- Tokuyama & Nakano (1995), Efficient algorithms for the Hitchcock transportation problem, SIAM J. Comput. 24, 563–578, doi:10.1137/S0097539792236729.
- Tseng & Ferziger (2001), Mixing and available potential energy in stratified flows, Phys. Fluids 13, 1281–1293.
- Winters, Lombard, Riley & D'Asaro (1995), Available potential energy and mixing in density-stratified fluids, JFM 289, 115–128, doi:10.1017/S002211209500125X.
- Winters & D'Asaro (1996), Diascalar flux and the rate of fluid mixing, JFM 317, 179–193.
- Wong, Tailleux & Gray (2016), The computation of reference state and APE production by diabatic processes in an idealized tropical cyclone, QJRMS 142, 2646–2657.
- Zadeh (1973), A bad network problem for the simplex method and other minimum cost flow algorithms, Math. Programming 5, 255–266, doi:10.1007/BF01580132.